Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

ISWE406P – Agile Development Process and Devops Lab

Assessment - 4

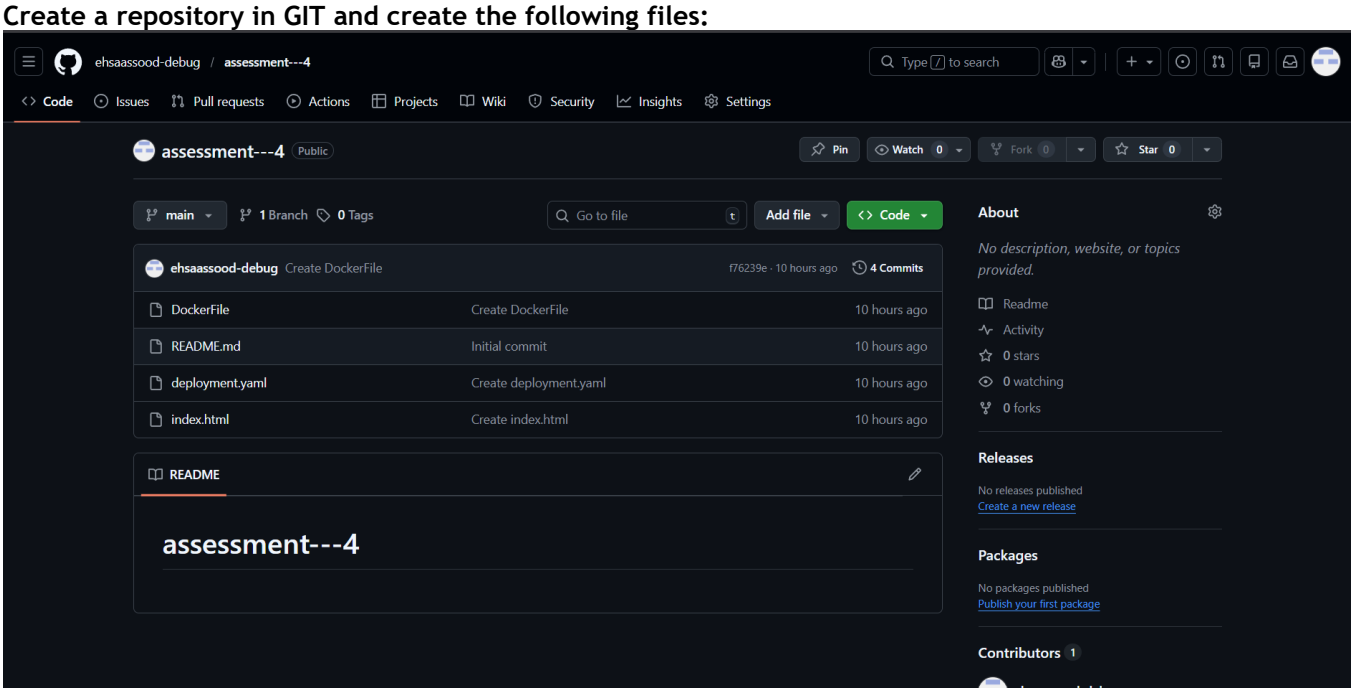
Submitted By :

Name : Ehsaas Sood

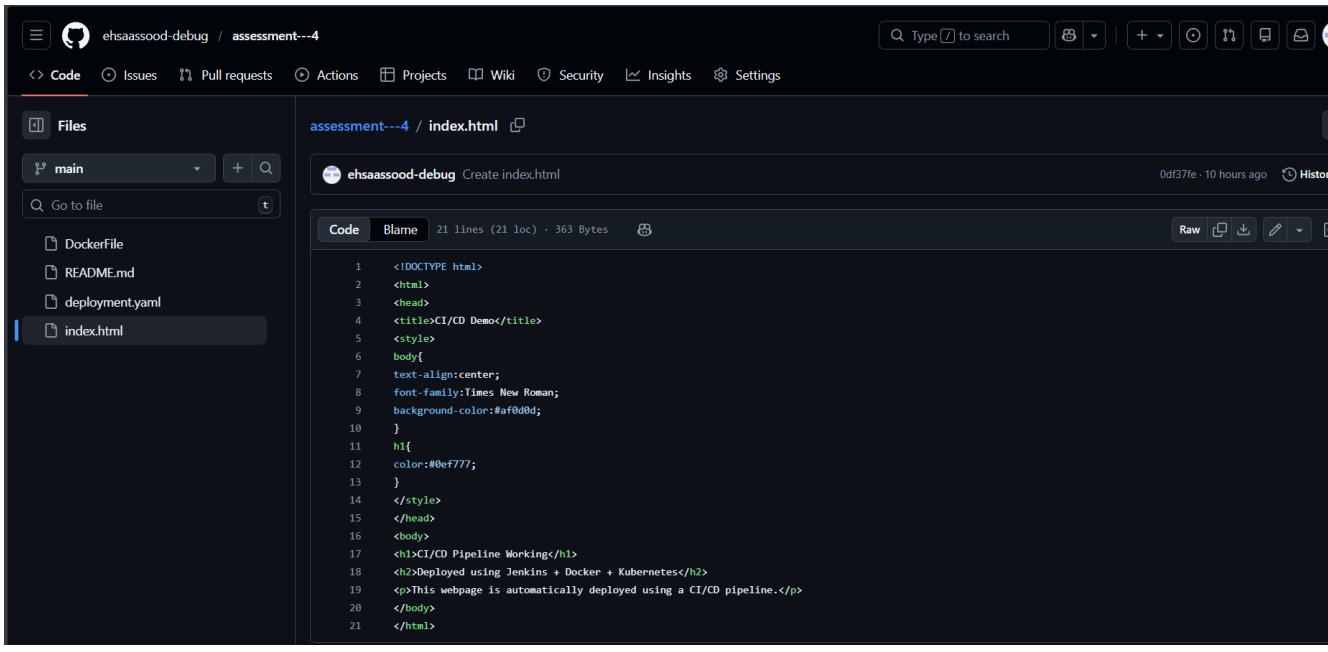
Register Number : 23MIS0074

Slot : L3 + L4

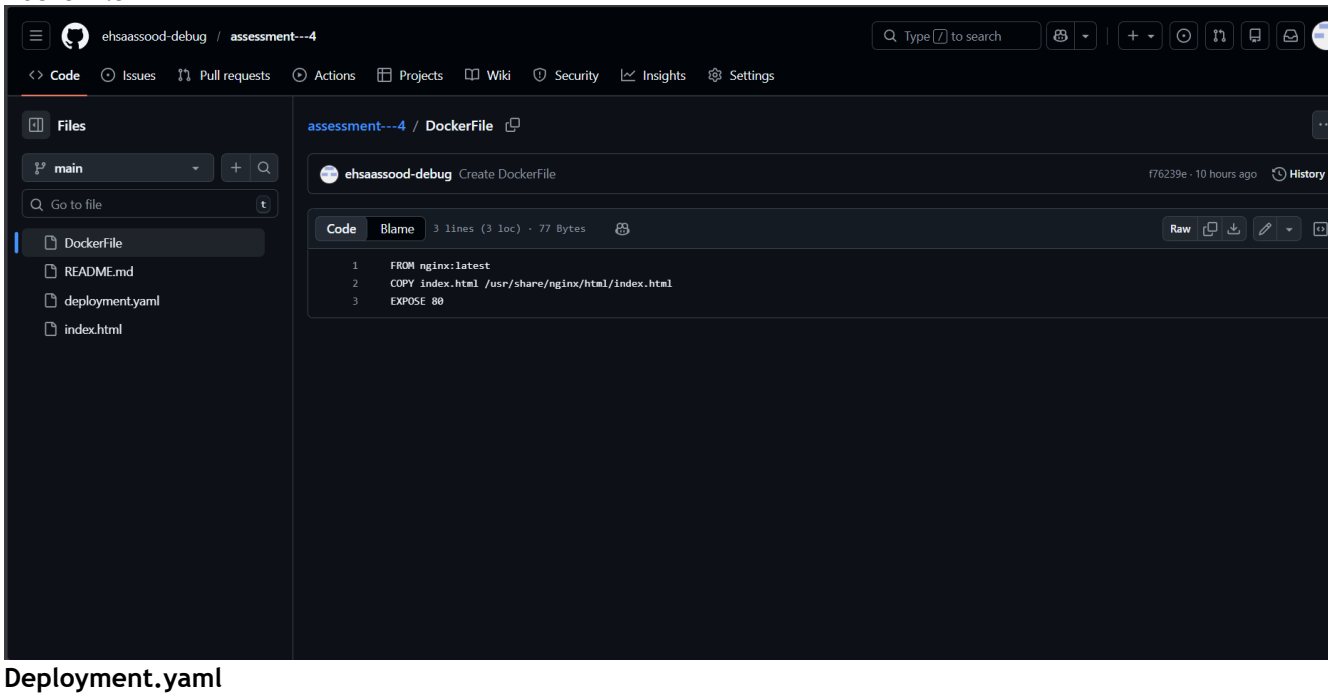
1. **Create a repository in GIT and create the following files:**



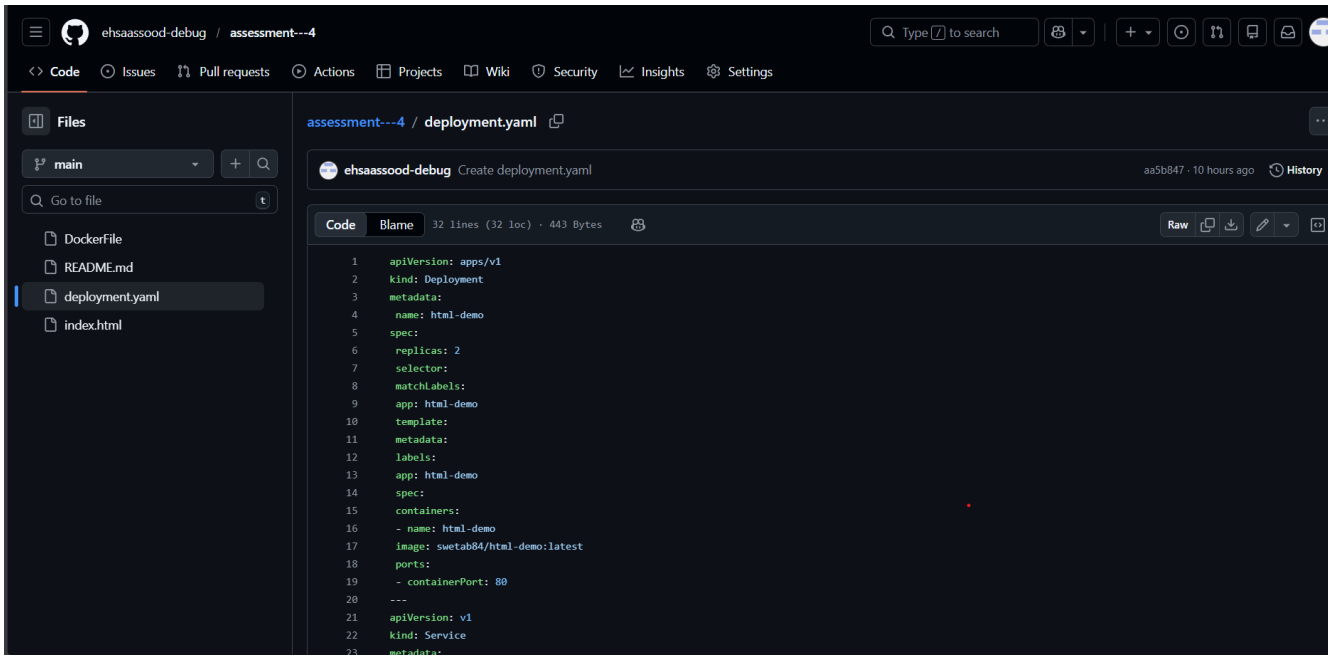
- **index.html**



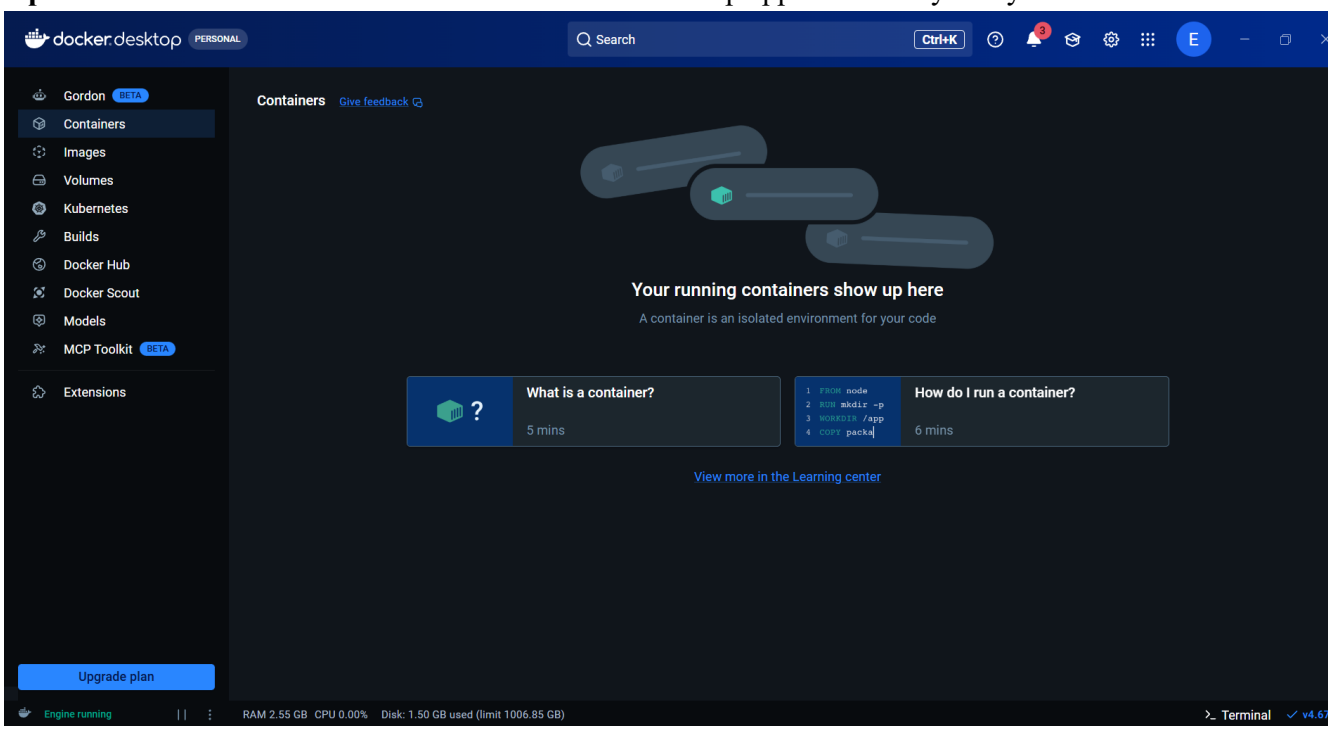
- Dockerfile



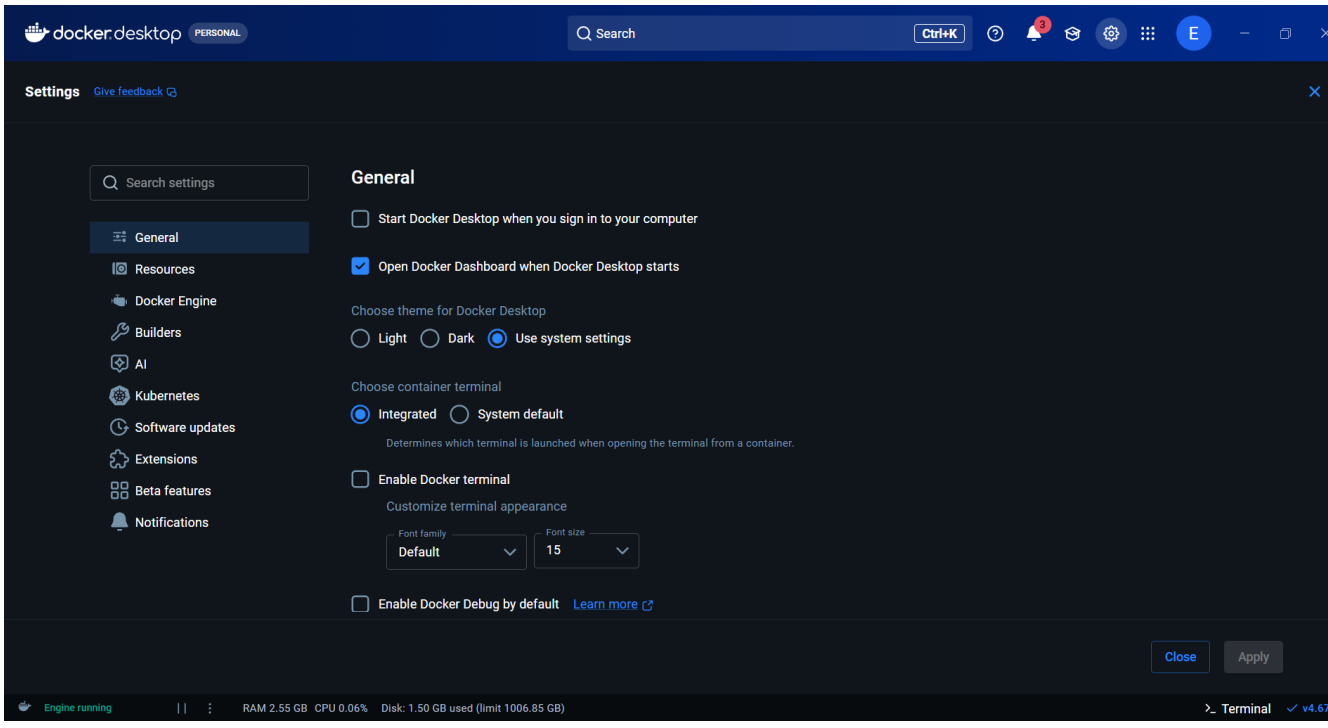
- **Deployment.yaml**



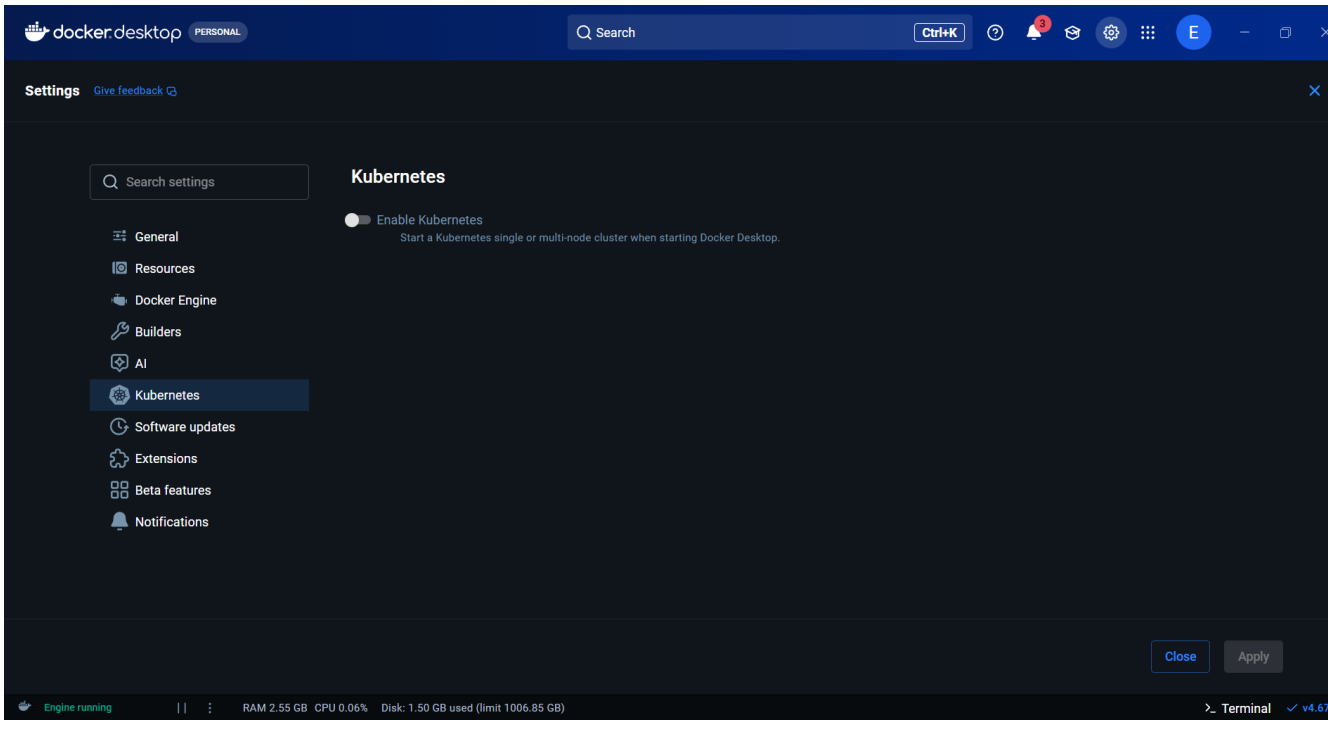
2. Enable Kubernetes
   - **Open the Docker Dashboard**: Launch the Docker Desktop application on your system.



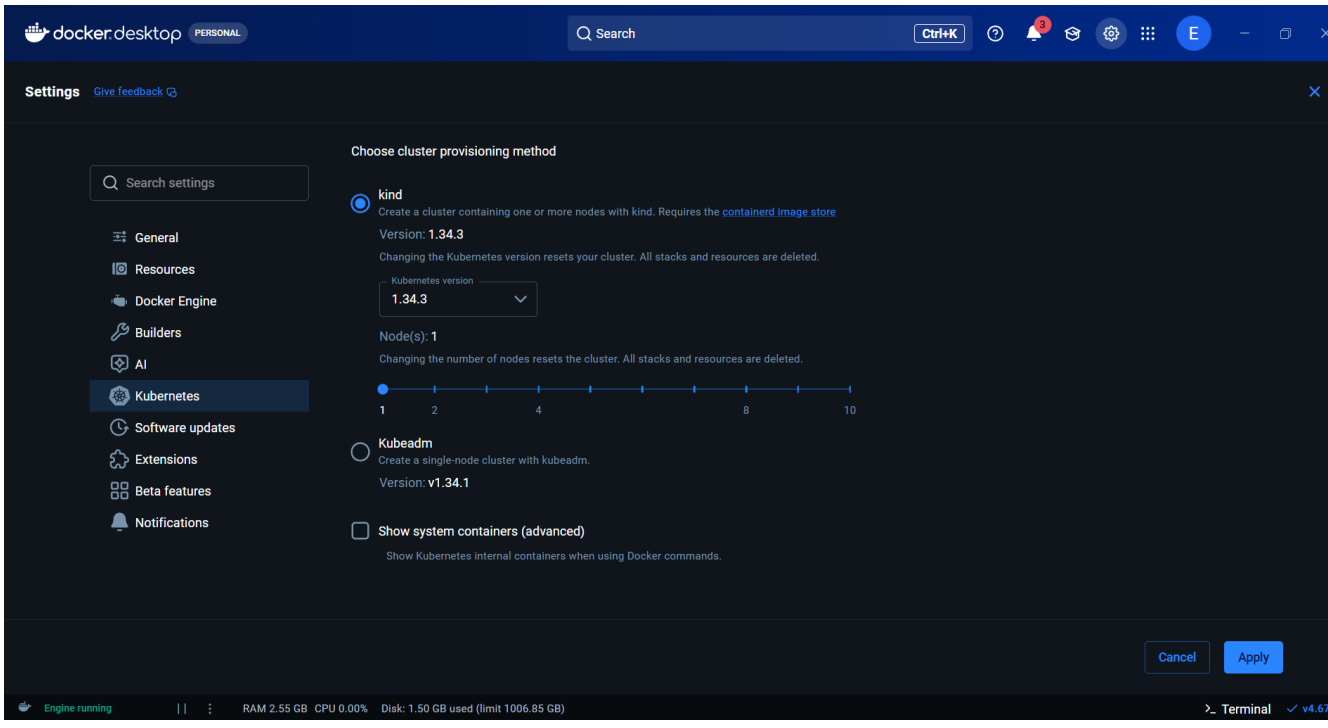
   - **Navigate to Settings**: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.



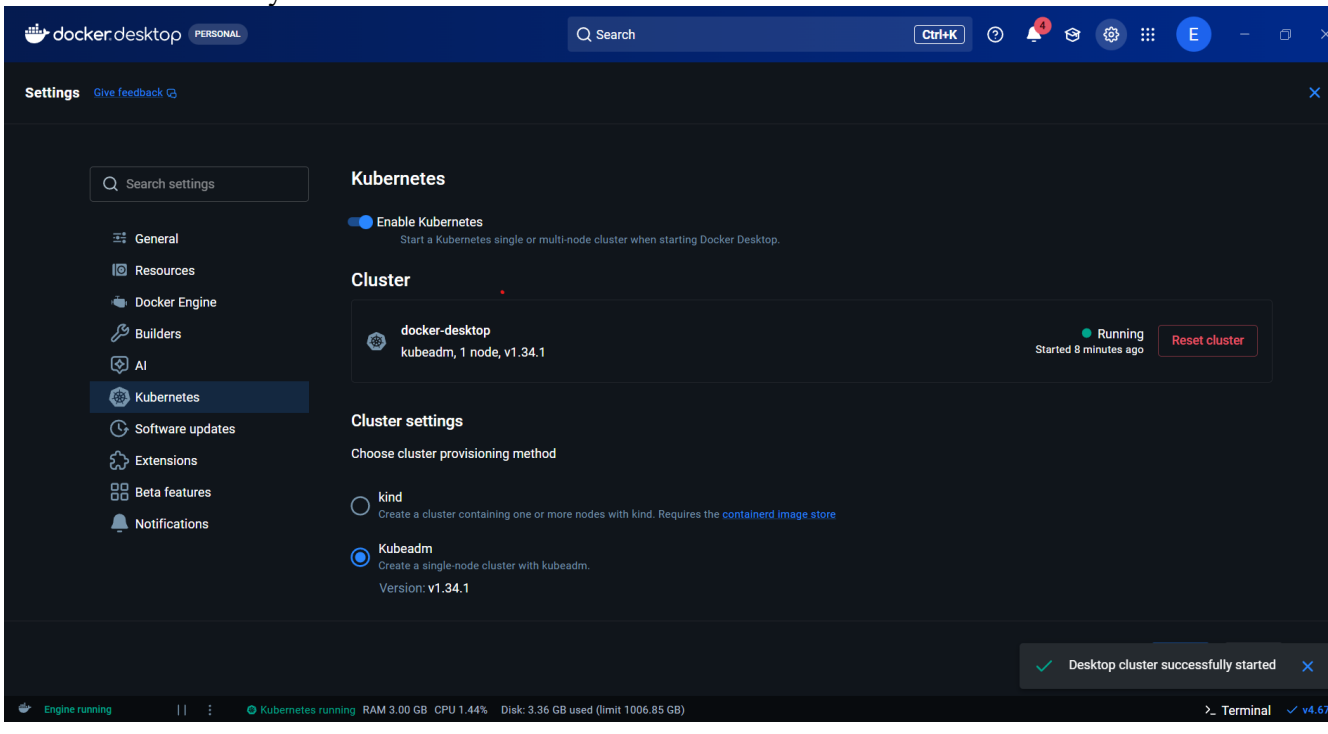
- **Select the Kubernetes Tab**: In the Settings menu, find and select the **Kubernetes** tab.



- **Enable Kubernetes**: Check the box labeled **Enable Kubernetes**.

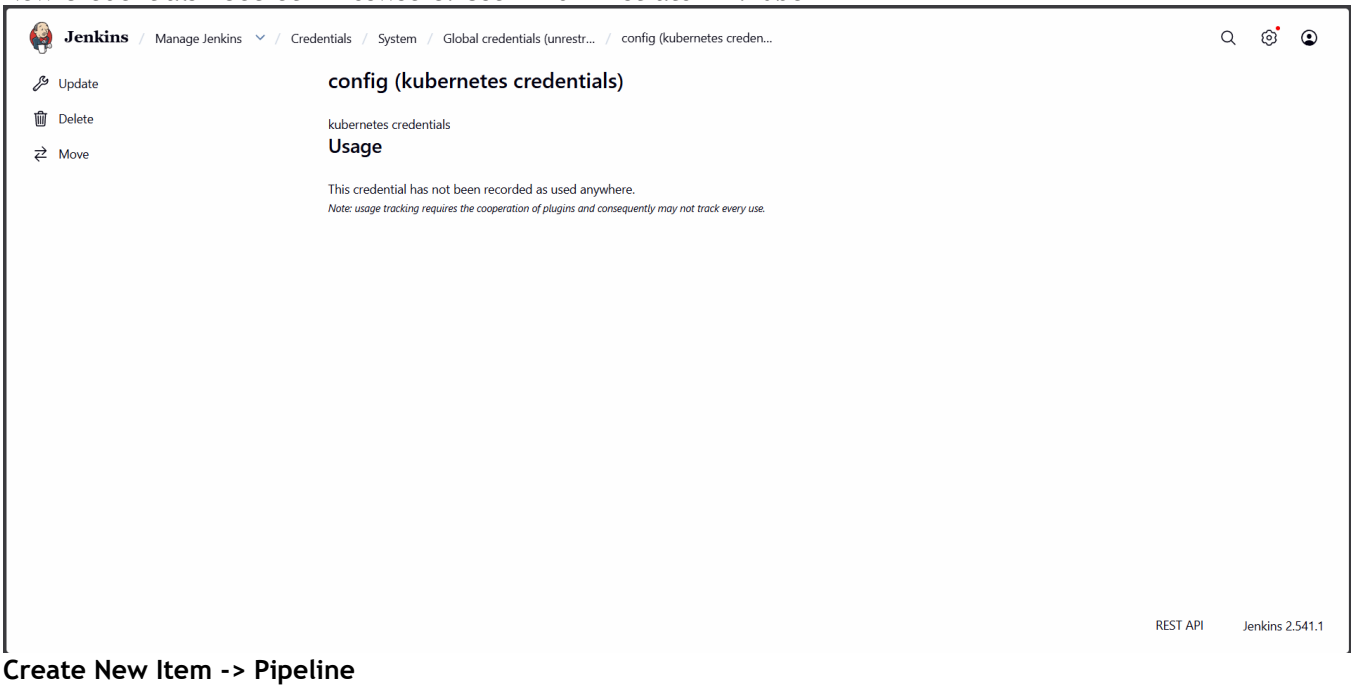


- **Apply & Restart**: Click the **Apply & Restart** button at the bottom of the settings window. Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time



3. Open Jenkins
   In Jenkins Setting --> Credentials New Credentials – User name: , password , ID:, Description

New Credentials – Secret – Broswse C: User->Administrator-->.kube



Create New Item -> Pipeline

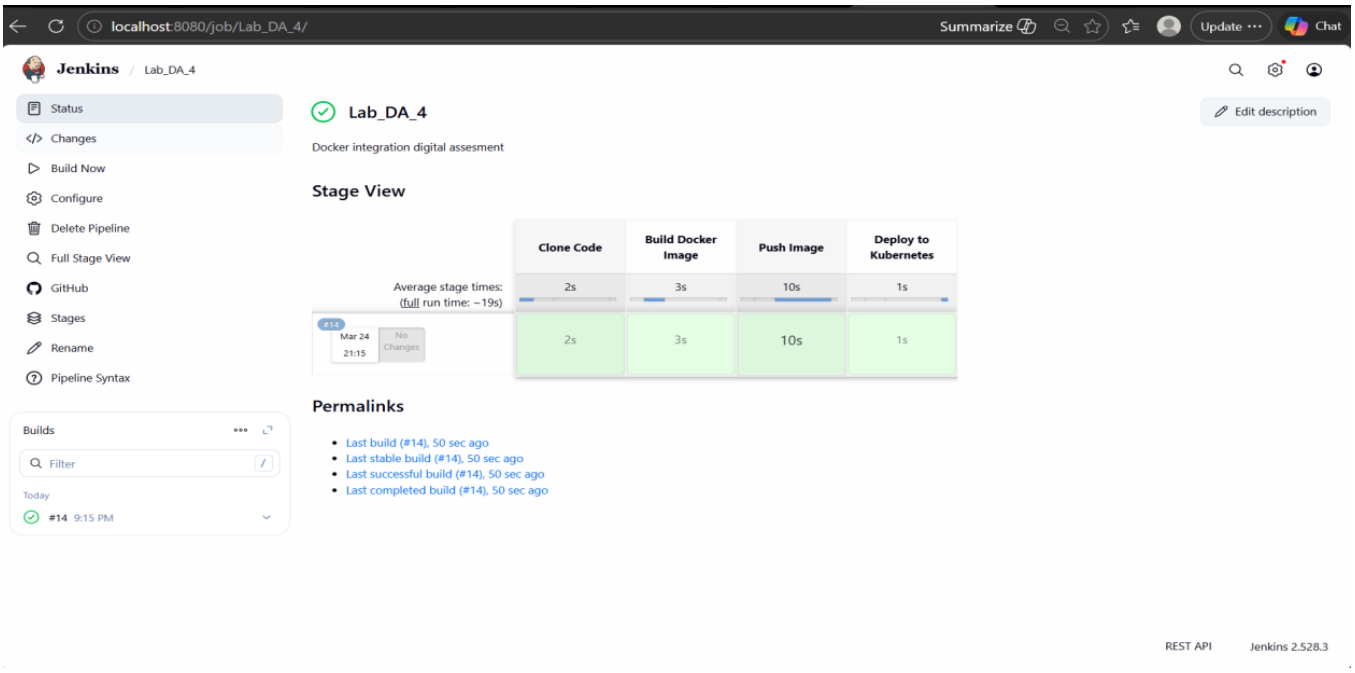
Paste the code in pipeline script

Update URL of GIT repository

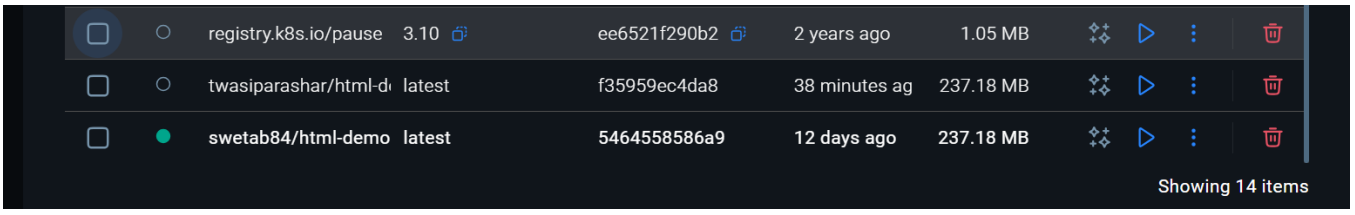
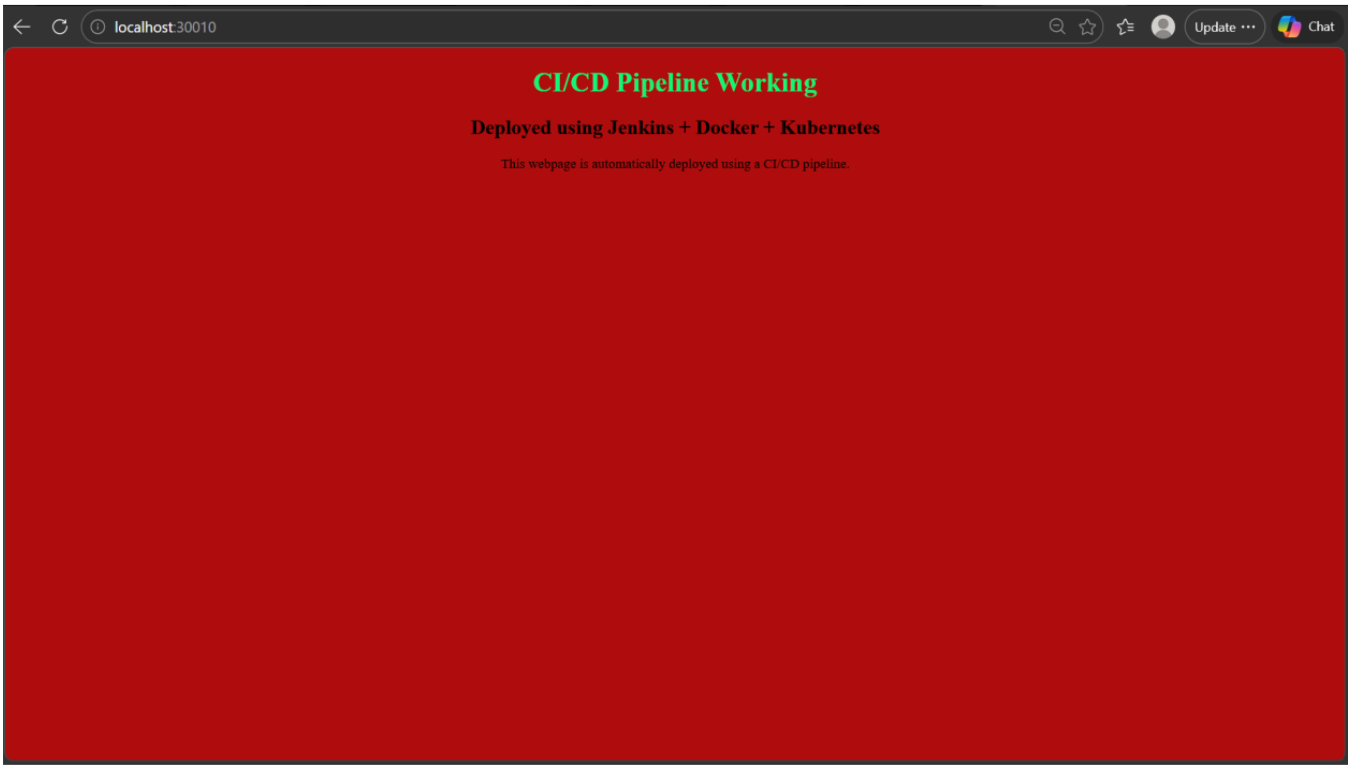
Make sure credentialsId: “dockerhub” same as mentioned in credentials

Make sure that the ID mentioned in credentials “kuberconfig” is same as in script

Run the Build



The same should be reflected in Dockerdesktop



Also the HTML page should open in the port mentioned in deployment.yaml file: nodePort: 30010